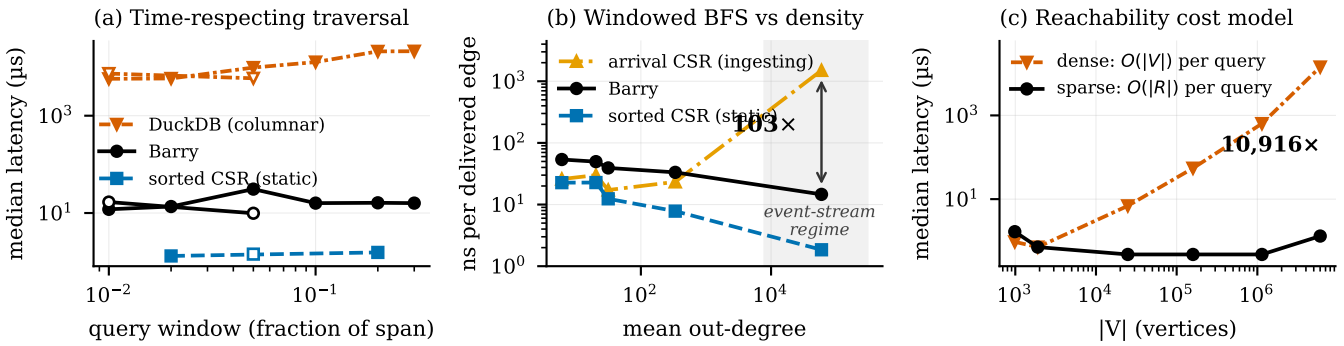
(a) Time-respecting traversal
median latency (µs)
$10^3$
$10^1$
DuckDB (columnar)
Barry
sorted CSR (static)
$10^{-2}$
$10^{-1}$
query window (fraction of span)
(b) Windowed BFS vs density
ns per delivered edge
$10^3$
$10^2$
$10^1$
$10^0$
arrival CSR (ingesting)
Barry
sorted CSR (static)
103×
event-stream regime
$10^2$
$10^4$
mean out-degree
(c) Reachability cost model
median latency (µs)
$10^3$
$10^1$
dense: $O(|V|)$ per query
sparse: $O(|R|)$ per query
10,916×
$10^3$
$10^4$
$10^5$
$10^6$
|V| (vertices)